

02_1_descriptive_inference_stats

August 26, 2025

1 Descriptive and Inference Statistics

The descriptive statistics summarizes the measures of central tendency, distribution, and the range of data.

I will use descriptive statistics to summarize - - Distribution and range of data - Central Limit Theorem & sampling - Bootstrapping - Apply the 'Plug-in principle' to complement CLT inferences - Confidence intervals - Hypothesis testing - Chi-squared tests for independence and categorical relationships - Analysis Of Variance (ANOVA) for multi-group comparison - check if there are statistical difference between means of three or more groups. - Visualization for interpretation

```
[1]: import pandas as pd
      from stats_utils import (
          describe_numeric_column, plot_distribution, plot_boxplot_by_target,
          clt_sampling_distribution, calculate_confidence_interval,
          bootstrap_confidence_interval, one_sample_ttest, two_sample_ttest,
          chi_squared_test, anova_test, calculate_z_score
      )
      df = pd.read_csv('../data/raw/telco-customer-churn-ibm.csv')
```

1.1 Descriptive Statistics for MonthlyCharges

```
[2]: # Monthly charges
      describe_numeric_column(df, 'MonthlyCharges')
```

Descriptive statistics for MonthlyCharges:

count	7043.000000
mean	64.761692
std	30.090047
min	18.250000
25%	35.500000
50%	70.350000
75%	89.850000
max	118.750000

Name: MonthlyCharges, dtype: float64

```
[2]: count    7043.000000
      mean      64.761692
```

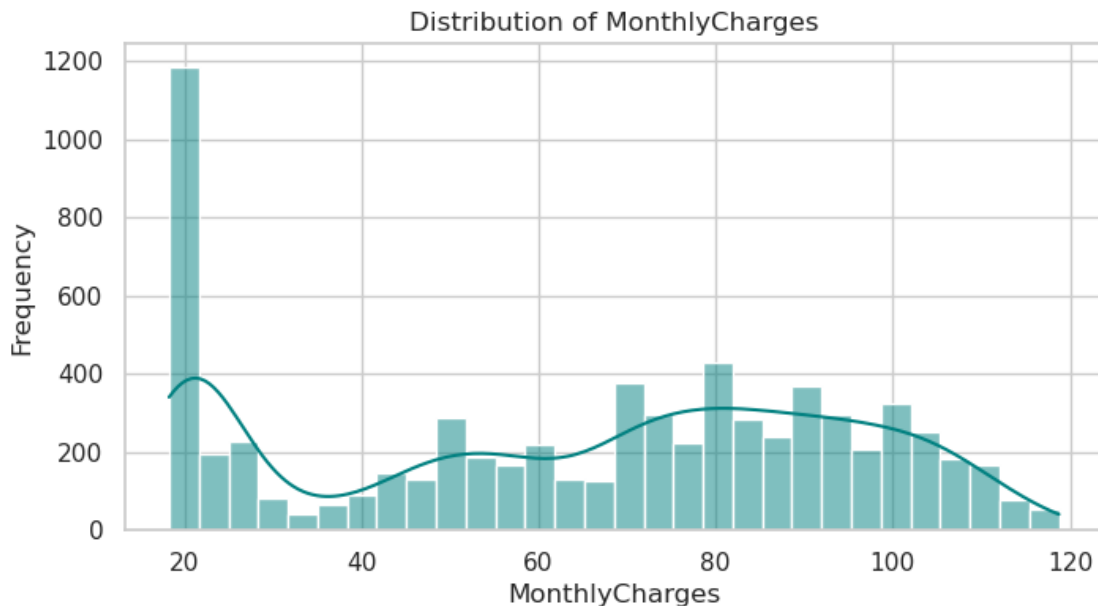
```
std      30.090047
min      18.250000
25%     35.500000
50%     70.350000
75%     89.850000
max     118.750000
Name: MonthlyCharges, dtype: float64
```

```
[3]: # Z-score
calculate_z_score(90, df['MonthlyCharges'].mean(), df['MonthlyCharges'].std())
```

Z-score: 0.839

```
[3]: 0.8387593232231267
```

```
[4]: plot_distribution(df, 'MonthlyCharges')
```



2 Confidence Interval for MonthlyCharges

2.1 95% Confidence Interval for Average MonthlyCharges

```
[5]: # Confidence interval
calculate_confidence_interval(df['MonthlyCharges'])
```

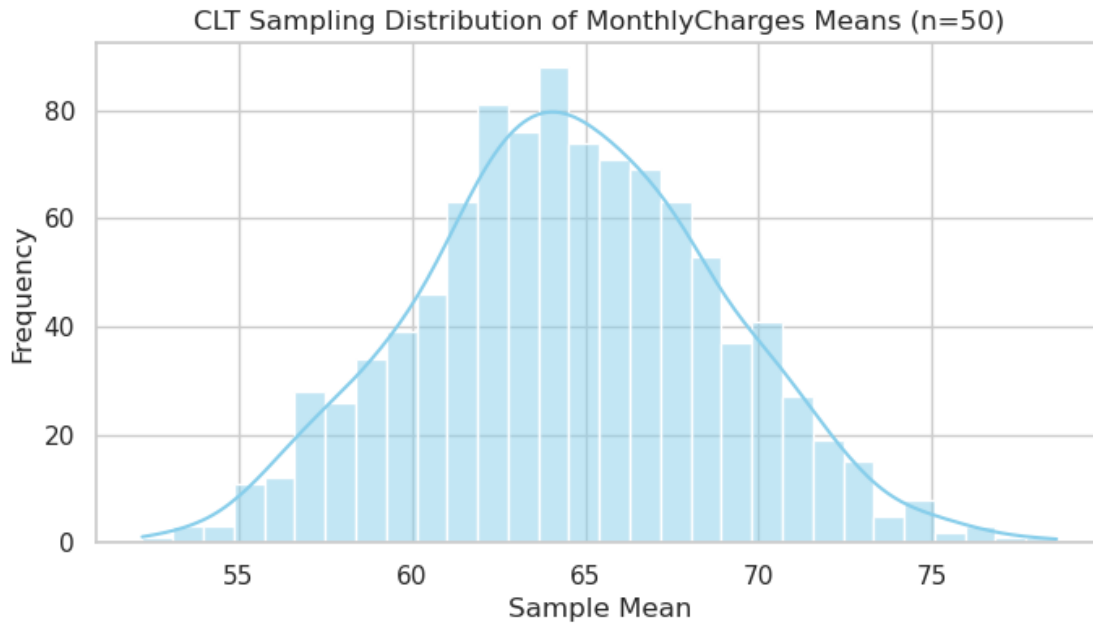
95% CI for mean: (64.06, 65.46)

```
[5]: (64.05883579770197, 65.46454912349638)
```

2.2 Central Limit Theorem: Sampling Distribution of the Mean

- the CLT for MonthlyCharges follows a normal distribution

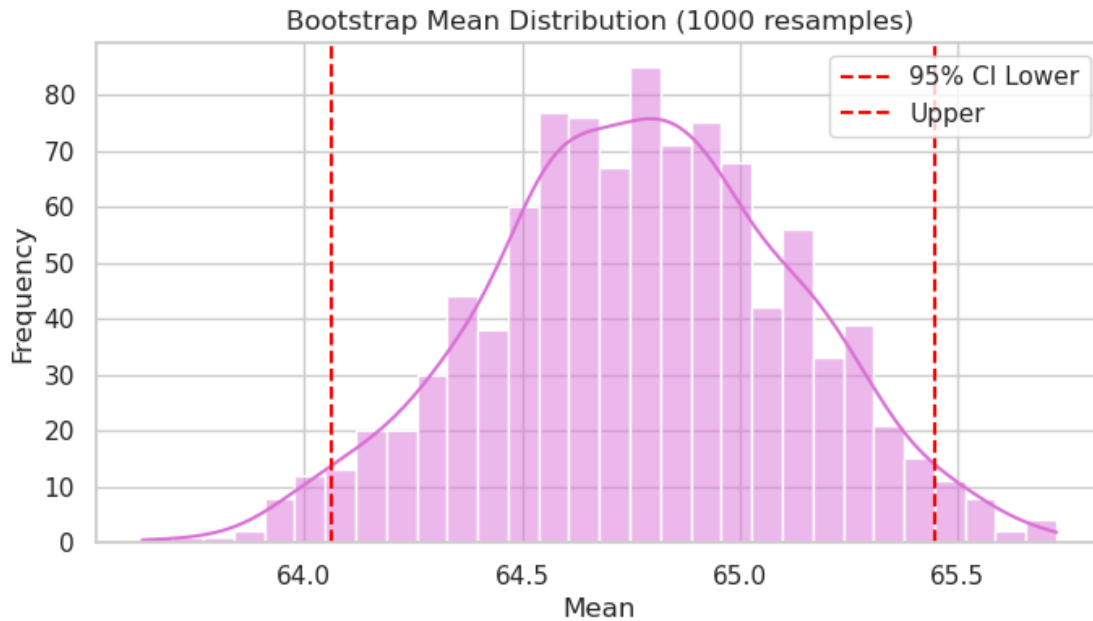
```
[6]: clt_sampling_distribution(df, 'MonthlyCharges')
```



3 Bootstrap Confidence Interval

3.1 Bootstrap Confidence Interval for Mean MonthlyCharges

```
[7]: bootstrap_confidence_interval(df['MonthlyCharges'])
```



95% bootstrap CI: (64.06, 65.44)

[7]: (64.06028592219224, 65.44477956836575)

3.2 Hypothesis Testing

```
[8]: # Hypothesis tests
one_sample_ttest(df['MonthlyCharges'], popmean=65)
```

One-sample t-test: t=-0.665, p=0.506

[8]: (-0.664651147898189, 0.5062954103387203)

```
[9]: # Churn vs not churned
churned = df[df['Churn']==1]['MonthlyCharges']
not_churned = df[df['Churn']==0]['MonthlyCharges']
two_sample_ttest(churned, not_churned)
```

Two-sample t-test: t=nan, p=nan

[9]: (nan, nan)

3.3 Chi-squared Test: Is Churn independent of Contract Type?

```
[10]: # Chi-squared test
table = pd.crosstab(df['Churn'], df['Contract'])
chi_squared_test(table)
```

Chi-squared test: chi2=1184.597, dof=2, p=0.000

```
[10]: (1184.5965720837926,  
       5.863038300673391e-258,  
       2,  
       array([[2846.69175067, 1082.11018032, 1245.198069 ],  
              [1028.30824933, 390.88981968, 449.801931 ]]))
```

3.4 ANOVA: Compare MonthlyCharges across Multiple Contract Types

```
[14]: # ANOVA (e.g., MonthlyCharges by contract type)  
groups = [df[df['Contract']==c]['MonthlyCharges'] for c in df['Contract'].  
          ↪unique()]  
anova_test(*groups)
```

ANOVA: F=20.828, p=0.000

```
[14]: (20.828045474730274, 9.575270975935273e-10)
```